



MonoTDF: Temporal deep feature learning for generalizable monocular 3D object detection

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ABSTRACT

Monocular 3D object detection has gained significant attention due to its cost-effectiveness and practicality in real-world applications. However, existing monocular methods often struggle with depth estimation and spatial consistency, limiting their accuracy in complex environments. In this work, we introduce a Temporal Deep Feature Learning framework, which enhances monocular 3D object detection by integrating temporal features across sequential frames. Our approach leverages a novel deep feature auxiliary module based on convolutional recurrent structures, effectively capturing spatiotemporal information to improve depth perception and detection robustness. The proposed module is model-agnostic and can be seamlessly integrated into various existing monocular detection frameworks. Extensive experiments across multiple state-of-the-art monocular 3D object detection models demonstrate consistent performance improvements, particularly in detecting small or partially occluded objects. Our results highlight the effectiveness and generalizability of the proposed approach, making it a promising solution for real-world autonomous perception systems. The source code of this work is at: <https://github.com/Shuray36/MonoTDF-Temporal-Deep-Feature-Learning-for-Generalizable-Monocular-3D-Object-Detection>.

1. Introduction

Object detection is a crucial task in computer vision, particularly for applications in autonomous driving, where systems must accurately interpret and interact with their environment. For autonomous vehicles, reliable object detection and precise spatial measurements from visual data are critical for safe navigation and decision-making. Traditional 2D object detection methods, which relied heavily on hand-crafted features [1,2], were limited in addressing the need for spatial accuracy in complex, dynamic environments. However, deep learning techniques have significantly improved 2D object detection, making it more robust and applicable in systems like Advanced Driver Assistance Systems (ADAS) [3,4].

In autonomous driving, accurate 3D object detection is essential for progression beyond Level 2 automation, as defined by the SAE J3016 standard [5]. Unlike 2D object detection, 3D object detection provides crucial information such as object location, size, orientation, and depth, which are indispensable for navigating complex environments [6].

While combining sensors such as LiDAR and cameras has improved 3D detection [7–9], sensor fusion remains computationally expensive and challenging for real-time applications. Models like RI-Fusion [10] and TFIENet [11] have demonstrated the potential of this fusion but also highlight the need for scalable, cost-effective solutions when multi-modal sensor data is unavailable.

Although monocular 3D object detectors have shown promising progress, they still struggle with small or partially occluded objects due to limited spatial cues available from a single frame. We hypothesize that incorporating temporal feature aggregation can enhance monocular 3D detection by leveraging motion continuity and inter-frame context, which are especially valuable for objects that are small, distant, or occluded. To address these challenges, we propose MonoTDF, a Temporal Deep Feature Auxiliary Module designed to enhance monocular 3D object detection. By incorporating temporal features from sequential frames, MonoTDF improves spatial measurement accuracy, resulting in more robust and precise object detection. This method uses convolutional recurrent neural network (RNN) structures, such as ConvLSTM

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and ConvGRU, to capture both spatial and temporal dependencies. We evaluate MonoTDF on baseline models like VisualDet3D [12,13] and Monodle [14], demonstrating significant improvements in detection accuracy, particularly for small and partially occluded objects. Our experiments on the KITTI dataset [15] highlight the effectiveness of our approach in enhancing detection performance, making it a promising solution for monocular-based autonomous perception systems.

To summarize, the key contributions of this work are as follows: 1). We design a lightweight, model-agnostic Temporal Deep Feature module based on convolutional recurrent units (ConvLSTM / ConvGRU) to exploit inter-frame temporal correlations for monocular 3D object detection. 2). The TDF module can be seamlessly integrated into existing monocular 3D detectors with minimal architectural modifications. We investigate multiple insertion points (after backbone, in neck, before prediction head) and identify the most effective configuration. 3). We conduct ablation studies on different temporal models (ConvRNN, ConvGRU, ConvLSTM) and varying sequence lengths to understand the trade-offs between temporal depth, efficiency, and accuracy. 4). We evaluate the proposed approach on KITTI and nuScenes, which are major benchmarks commonly used nowadays, and validate its generalizability by integrating TDF into multiple state-of-the-art monocular 3D detectors (e.g., VisualDet3D and Monodle). 5). The proposed TDF module consistently enhances 3D and BEV detection performance across object categories and datasets, without requiring additional depth, LiDAR, or stereo information. The remainder of this paper is organized as follows: Section II reviews relevant deep learning architectures for monocular 3D object detection. Section III details the design and integration of our temporal auxiliary module. Section IV presents the experimental setup and evaluation of the proposed model, and Section V concludes with key findings and suggestions for future research.

2. Related work

Monocular 3D object detection is gaining attention for its cost-effectiveness in applications like autonomous driving [16] and robotics. However, depth estimation and spatial inconsistency remain key challenges, especially in complex scenes. Recent studies address these issues by incorporating both spatial and temporal cues.

2.1. Monocular 3D object detection

Early monocular 3D detection methods, such as Mono3D [17] and Deep3DBox [18], used geometric constraints from 3D box projections to estimate depth and location, but depended on priors or external data, limiting generalization. GS3D [19] and MonoRCNN [20] added semantic cues but struggled with occlusions and scale variation. Models like MonoFlex [21] and YOLOStereo3D [12] used stereo cues for better depth but required extra sensors or complex features. Some CNN-based methods directly regress 3D attributes [22], yet monocular vision remains ill-posed, causing depth uncertainty.

VisualDet3D addresses this by introducing ground-aware convolution and anchor filtering, which leverage ground priors and exclude implausible 3D positions, reducing localization errors. Despite advances, monocular depth estimation remains under-constrained. Pseudo-LiDAR [23] and Pseudo-LiDAR++ [24] generate point clouds from images for 3D detection, but still suffer from depth artifacts and long-range errors compared to LiDAR.

2.2. Temporal and spatial-temporal learning

To enhance robustness and depth estimation, recent studies have leveraged temporal information from sequential frames. Temporal consistency offers contextual cues for moving objects, reducing depth uncertainty. Models like Deep MANTA [25] and DD3D [26] integrate multi-frame features to refine depth and pose, improving accuracy under occlusion and motion blur. Mono3D-PLiDAR [27] further

incorporates temporal cues for better 3D box prediction but still relies on external depth networks. Recurrent neural networks (RNNs) and temporal aggregation methods have been used to maintain spatial continuity across video frames, aiding detection of dynamic or occluded objects. However, they face challenges in training on long sequences and coping with noise or abrupt motions.

2.3. Temporal deep feature learning for 3D object detection

Our method, MonoTDF, introduces a deep feature learning framework that models temporal relationships in monocular 3D detection. Using convolutional recurrent structures, it captures both short- and long-term dependencies to enhance depth perception and robustness in dynamic scenes. Unlike prior temporal models, MonoTDF is framework-agnostic and easily integrates into existing systems.

While multimodal approaches like MV3D [28] and ContFuse [29] fuse camera and LiDAR data via various strategies, they face challenges with semantic misalignment and high complexity. In contrast, MonoTDF leverages temporal consistency from monocular videos, offering a simpler, sensor-free alternative. Additionally, geometry-supervised models such as LIGA-Stereo [30] and 3D-CVF [31] highlight the benefits of structural cues. By combining temporal learning with geometry-aware losses, MonoTDF improves detection, especially for small or occluded objects.

3. Methodology

This section presents the motivation behind our approach, describes the proposed architectural framework incorporating modified convolutional RNN structures, and outlines the inference process of the model.

3.1. Motivation

Monocular 3D object detection remains challenging due to the absence of explicit depth, making it less reliable than LiDAR- or stereo-based methods. Despite progress in deep learning, monocular models still face depth ambiguity, occlusions, and limited spatial cues. Studies show that temporal information from sequential frames enhances 3D detection by capturing motion and structural consistency [32,33]. Building on this, we explore deep temporal features to improve robustness and generalization. We propose incorporating convolutional RNNs, such as ConvLSTM and ConvGRU, into monocular detection networks. Unlike traditional RNNs, these structures process high-dimensional feature maps, allowing effective fusion of spatial and temporal cues.

3.2. Architectural framework

Our approach extends conventional RNN architectures by embedding 2D convolutional layers within ConvLSTM and ConvGRU cells. As illustrated in Fig. 1, the hidden state from the previous time step h_{t-1} and the current input feature map x_t undergo a convolutional transformation, extracting spatial and temporal representations before feature concatenation. This process enhances the model's ability to track objects over time, improving detection robustness. The ConvLSTM

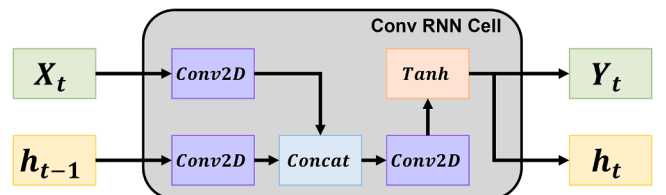


Fig. 1. The structure of convolutional RNN (ConvRNN) Cell.

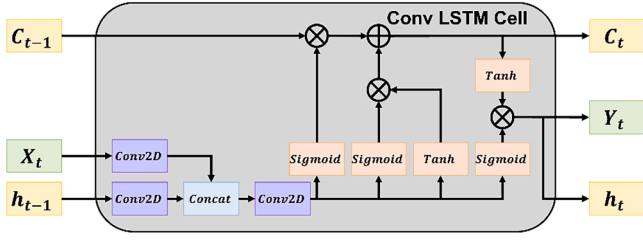


Fig. 2. The structure of convolutional LSTM (ConvLSTM) Cell.

structure retains forgetting, input, and output gates similar to the standard LSTM, while incorporating 2D convolutions for feature extraction, as shown in Fig. 2, and the ConvGRU structure adopts a more streamlined design, with convolutional layers replacing linear transformations in the GRU, allowing for efficient high-dimensional feature processing, as shown in Fig. 3.

Our convolutional RNN structures are integrated as auxiliary temporal modules within monocular 3D object detection pipelines. The optimal placement is after the neck structure of the model, where multi-scale features are aggregated. This placement ensures that the network captures both local and global contextual cues before temporal refinement. The integration strategy is visualized in Fig. 4.

By incorporating convolutional recurrent modules, our framework enhances spatial measurements, including depth, boundaries, and positional relationships. These refinements are crucial for improving accuracy in complex urban environments.

3.3. Inference flow

Unlike conventional monocular 3D object detection models that process individual frames independently, our model leverages multiple consecutive frames as input, enabling it to capture temporal dependencies. The inference process begins with feature extraction, where the backbone extracts spatial features from the input image sequence. Next, the extracted feature maps will serve as the input X_t , undergoing processing through convolutional RNN structures. Subsequent to the processing, the output Y_t , cell state C_t , and hidden state h_t can be obtained. The cell state C_t and hidden state h_t will then combine with the extracted feature maps of the subsequent time step, and become the input of the convolutional recurrent neural network (RNN) structure of the next time step. This hierarchical process enriches the entire framework for obtaining motion- and depth-aware information. Finally, the detection head generates the 3D object bounding boxes and depth estimates by utilizing both spatial and temporal cues.

Once the initial detection is made, only one additional frame is required for each subsequent prediction, ensuring computational efficiency. This design allows our method to remain cost-effective while significantly improving performance compared to single-frame monocular detection models. The effect of varying the number of observed frames on model performance is demonstrated in Fig. 5.

4. Experiments

This section presents the details of the experimental dataset, the 3D detection indicators, the effectiveness comparisons among different

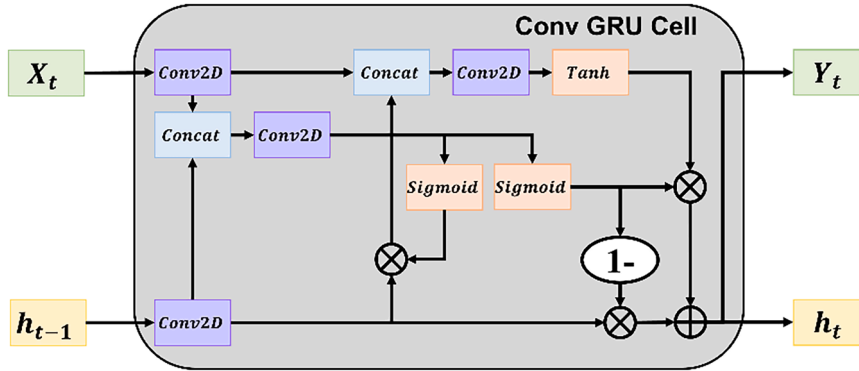


Fig. 3. The structure of convolutional GRU (ConvGRU) Cell.

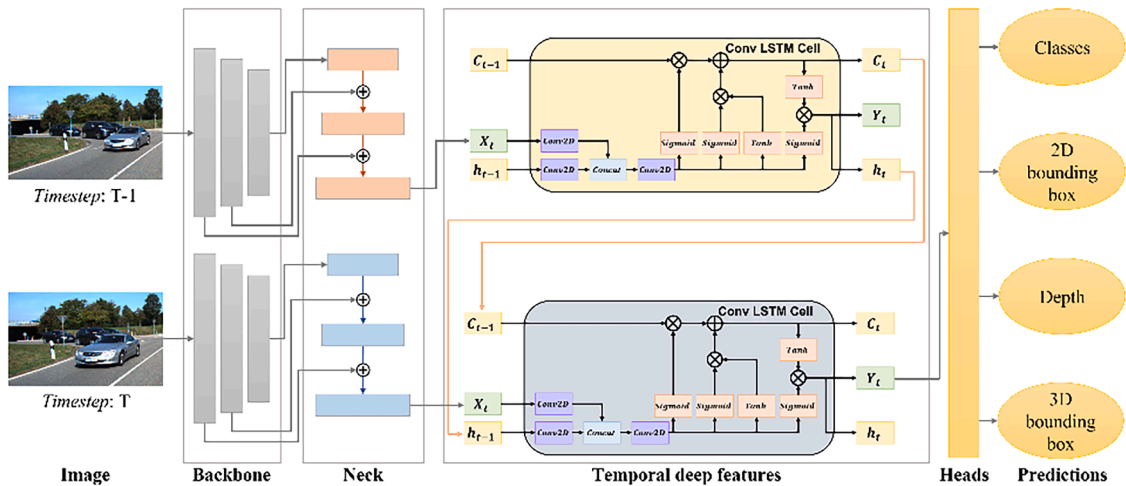


Fig. 4. The conceptual architectures of 3D object detection model with the convolutional LSTM Cell.

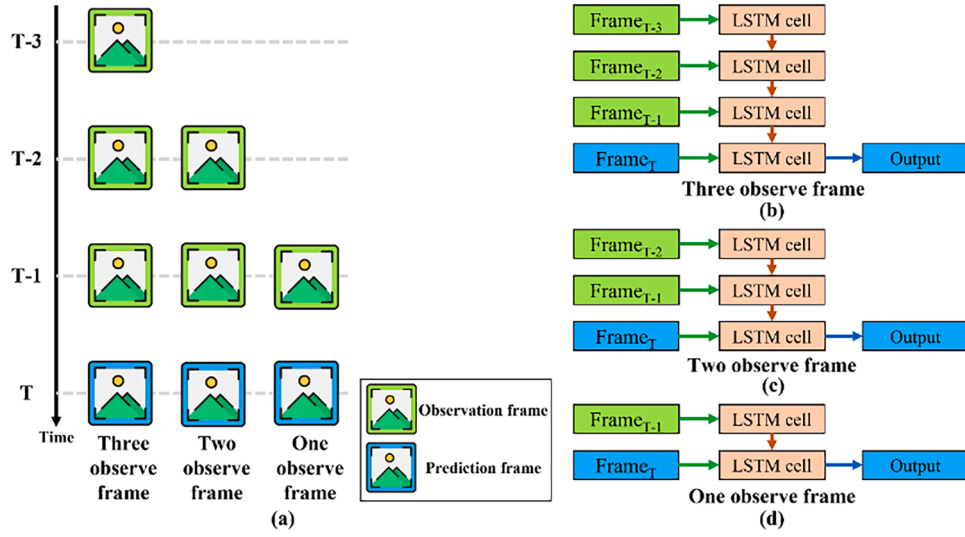


Fig. 5. The diagram illustrates the varying structures for different numbers of observed image frame.

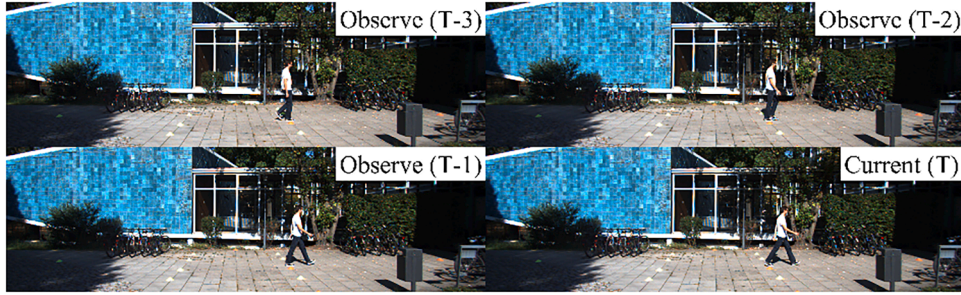


Fig. 6. Sample images of KITTI sequential data.

proposed architectures, optimization of the auxiliary module, and comparative performance analysis.

4.1. Dataset

The KITTI dataset, a standard benchmark for monocular 3D object detection, captures diverse driving scenarios with varying occlusion and

truncation. It defines three difficulty levels, i.e. Easy, Moderate, and Hard, based on object size and visibility. KITTI includes labeled and unlabeled images recorded at 10 Hz, with each labeled frame accompanied by three preceding frames (T-3 to T-1), as shown in Fig. 6. This temporal structure supports motion and sequence feature extraction, crucial for enhancing detection. Our experiments focused on cars, pedestrians, and cyclists.



Fig. 7. Images under different weather conditions in nuScenes.



Fig. 8. Sample images of nuScenes sequential data.

Table 1

Effectiveness comparison result between different proposed cell structure.

KITTI Car	AP _{2D} (0.7) ↑			AP _{3D} (0.7) ↑			AP _{3D} (0.5) ↑		
	E	M	H	E	M	H	E	M	H
VisualDet3D [13]	97.30	84.54	64.65	19.43	13.60	10.82	55.49	39.03	30.86
ConvRNN Cell	97.28	84.55	64.66	21.77	15.41	11.85	56.21	39.59	31.36
	(−0.02)	(+0.01)	(+0.01)	(+2.34)	(+1.81)	(+1.03)	(+0.72)	(+0.56)	(+0.5)
ConvLSTM Cell	97.33	84.49	67	21.24	15.78	12.07	59.13	41.71	32.02
	(+0.03)	(−0.05)	(+2.35)	(+1.81)	(+2.18)	(+1.25)	(+3.64)	(+2.68)	(+1.16)
ConvGRU Cell	97.27	86.92	67.06	20.89	14.66	11.74	57.32	41.44	31.82
	(−0.03)	(+2.38)	(+2.41)	(+1.46)	(+1.06)	(+0.92)	(+1.83)	(+2.41)	(+0.96)

Table 2

Effectiveness comparison between different position for proposed auxiliary module.

KITTI Car	AP _{2D} (0.7) ↑			AP _{3D} (0.7) ↑			AP _{3D} (0.5) ↑		
	E	M	H	E	M	H	E	M	H
VisualDet3D [13]	97.30	84.54	64.65	19.43	13.60	10.82	55.49	39.03	30.86
After backbone	87.39	72.21	54.78	17.09	11.25	8.58	51.78	35.05	27.01
	(−9.91)	(−12.3)	(−9.87)	(−2.34)	(−2.35)	(−2.24)	(−3.71)	(−3.98)	(−3.85)
In neck	94.50	76.99	59.58	18.58	12.56	9.81	52.75	36.42	28.53
	(−2.8)	(−7.55)	(−5.07)	(−0.85)	(−1.04)	(−1.01)	(−2.74)	(−2.61)	(−2.33)
Before head	97.33	84.49	67	21.24	15.78	12.07	59.13	41.71	32.02
	(+0.03)	(−0.05)	(+2.35)	(+1.81)	(+2.18)	(+1.25)	(+3.64)	(+2.68)	(+1.16)

In addition to KITTI, this study also uses the nuScenes dataset [34], widely adopted in autonomous driving research. It includes data from six cameras, five RADARs, and one LiDAR, covering 1000 scenes under various conditions (e.g., day, night, rain), as shown in Fig. 7. Objects are labeled across 23 categories. nuScenes captures data at 15 FPS, assigning each labeled sample five associated frames (Fig. 8). Unlike KITTI, its data is sequentially organized, allowing for easier expansion of temporal observations when additional labels are added.

4.2. Training details

All experiments were conducted on a workstation with dual NVIDIA RTX A5000 GPUs (24GB each) and 48GB RAM, running Ubuntu 18.04.6. The models were implemented in Python 3.8 using PyTorch with CUDA 11.1 and mixed-precision (AMP) training for efficiency. Gradient clipping (max norm 5.0) was applied to stabilize recurrent units. We used the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay $= 1 \times 10^{-4}$),

Table 3

Effectiveness comparison between different number of observed image frames.

KITTI Car	AP _{2D} (0.7) ↑			AP _{3D} (0.7) ↑			AP _{3D} (0.5) ↑		
	E	M	H	E	M	H	E	M	H
VisualDet3D [13]	97.30	84.54	64.65	19.43	13.60	10.82	55.49	39.03	30.86
Three frames	97.21 (−0.09)	84.47 (−0.07)	64.62 (−0.03)	19.69 (+0.26)	14.23 (+0.63)	11.48 (+0.66)	56.13 (+0.64)	39.67 (+0.64)	31.49 (+0.63)
Two frames	97.26 (−0.04)	84.56 (+0.02)	67.05 (+2.4)	20.01 (+0.58)	14.38 (+0.78)	11.64 (+0.82)	58.24 (+2.75)	41.76 (+2.73)	32.00 (+1.14)
One frame	97.22 (−0.08)	84.49 (−0.05)	67.00 (+2.35)	21.24 (+1.81)	15.78 (+2.18)	12.07 (+1.25)	59.13 (+3.64)	41.71 (+2.68)	32.02 (+1.16)

Table 4

Modified VisualDet3D performance of the car category on the KITTI test set.

	AP _{2D} (0.7) ↑			AP _{BEV} (0.7) ↑			AP _{3D} (0.7) ↑		
	E	M	H	E	M	H	E	M	H
VisualDet3D [13]	96.75	84.07	64.66	26.66	19.35	15.06	18.96	13.73	10.72
Ours	96.75 (+0.00)	84.07 (+0.00)	66.06 (+1.40)	28.48 (+1.82)	20.55 (+1.21)	16.12 (+1.06)	20.90 (+1.94)	15.27 (+1.54)	11.77 (+1.05)

Table 5

Modified VisualDet3D performance of the pedestrian category on the KITTI test set.

	AP _{2D} (0.5) ↑			AP _{BEV} (0.5) ↑			AP _{3D} (0.5) ↑		
	E	M	H	E	M	H	E	M	H
VisualDet3D [13]	55.98	46.22	39.29	8.39	6.71	5.09	7.44	5.83	4.64
Ours	58.43 (+2.45)	47.05 (+0.83)	40.14 (+0.84)	9.46 (+1.07)	7.52 (+0.81)	5.69 (+0.60)	8.31 (+0.87)	6.49 (+0.66)	5.14 (+0.50)

with a cosine annealing learning rate schedule starting at 1×10^{-4} and decaying to 1×10^{-6} , and a linear warm-up phase of 5 epochs (KITTI) or 2 (nuScenes).

Training ran for 80 epochs on KITTI and 24 on nuScenes, using per-GPU batch sizes of 8 and 6 respectively, and data parallelism. Each sample included frame T and its previous frame (T−1) for temporal context in the TDF module. Images were normalized with ImageNet statistics; backbones used ImageNet-pretrained weights. The TDF module was trained jointly with the base detector using the same loss functions as baseline models. For reproducibility, we used fixed random seeds {2023, 2024, 2025} and enabled deterministic settings; results are reported using seed 2024.

Inference speed and FLOPs were measured on a single RTX A5000 GPU with batch size 1. VisualDet3D ran at 28.7 FPS (105.8 GFLOPs), reduced to 27.2 FPS (108.6 GFLOPs) with TDF (+2.6 % overhead). Monodle dropped from 30.1 to 28.9 FPS, with FLOPs increasing from 98.4 to 101.0 (+2.7 %). These results show that TDF adds minimal computational cost while improving 3D and BEV accuracy. Evaluation followed official KITTI (3712 train / 3769 validation) and nuScenes (700 train / 150 validation) splits, using only monocular inputs. Standard metrics (AP_{2D}, AP_{BEV}, and AP_{3D}) and KITTI difficulty levels (Easy, Moderate, Hard) were applied throughout.

4.3. 3D Detection indicators

This study adopts three standard 3D object detection metrics: AP_{2D}, AP_{BEV}, and AP_{3D}. AP_{2D} evaluates 2D bounding box accuracy in image space, while AP_{BEV} measures detection from a top-down view, essential for autonomous driving. AP_{3D} assesses full 3D localization, considering object position, size, and orientation. Together, these metrics offer a comprehensive assessment of detection performance across spatial dimensions.

4.4. Effectiveness comparison of proposed architectures

To assess the proposed convolutional recurrent modules, we integrated them into the VisualDet3D baseline, which includes a spatial backbone, feature-processing neck, and a prediction head. Three architectures were tested: ConvRNN, ConvLSTM, and ConvGRU.

As shown in Table 1, ConvLSTM achieved the best overall performance on the KITTI dataset, evaluated using AP_{2D} and AP_{3D} at IoU thresholds of 70 % and 50 % across difficulty levels. It performed especially well in the Moderate and Hard categories, where monocular methods typically struggle, and remained competitive even when not the top performer, demonstrating robustness in complex scenarios.

While AP_{2D} improvements in Table 1 are minor (e.g., −0.02, +0.01), this aligns with the near-saturation of 2D detection and its low sensitivity to temporal modeling. In contrast, AP_{3D} gains are more substantial and consistent across all difficulty levels and IoU thresholds. All recurrent variants (ConvRNN, ConvLSTM, ConvGRU) outperformed the VisualDet3D baseline, with AP_{3D} increases over 1 point in Moderate and Hard cases at IoU = 0.7, and consistent improvements at IoU = 0.5. These patterns, observed across architectures, suggest that the benefits stem from the temporal modules rather than random variation, highlighting their stable contribution to depth-aware 3D localization.

4.5. Optimizing the auxiliary module placement

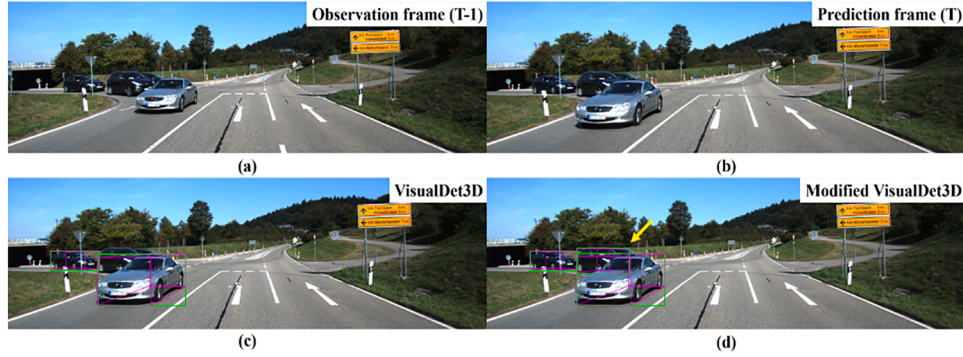
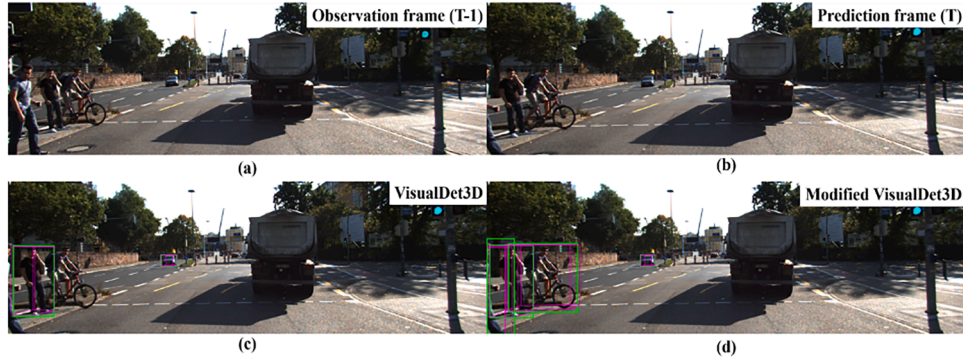
To identify the optimal placement of the ConvLSTM module, three configurations were tested: after the backbone, within the neck, and before the prediction head. As shown in Table 2, inserting the module before the head achieved the best results, likely due to its ability to enhance multi-scale spatial features prior to final prediction, improving localization and depth estimation.

Table 3 analyzes the impact of temporal window size, comparing one, two, and three previous frames. Using a single prior frame yielded the best balance of accuracy and efficiency. Although more frames

Table 6

Modified VisualDet3D performance of the cyclist category on the KITTI test set.

	$AP_{2D}(0.5) \uparrow$			$AP_{BEV}(0.5) \uparrow$			$AP_{3D}(0.5) \uparrow$		
	E	M	H	E	M	H	E	M	H
VisualDet3D [13]	53.09	32.25	30.43	3.59	1.98	2.00	3.04	1.72	1.65
Ours	54.61	33.81	31.67	4.46	2.77	2.70	3.95	2.32	2.36
	(+1.52)	(+1.56)	(+1.24)	(+0.87)	(+0.79)	(+0.70)	(+0.91)	(+0.60)	(+0.71)

**Fig. 9.** Visualization result of VisualDet3D and the modified VisualDet3D with the proposed convolutional LSTM Cell auxiliary module.**Fig. 10.** Visualization result of VisualDet3D and the modified VisualDet3D with the proposed convolutional LSTM Cell auxiliary module.

offered slight gains, they introduced diminishing returns and increased computational cost. Notably, performance degraded when using three frames, likely due to feature misalignment from object and camera motion. As ConvLSTM assumes temporal spatial consistency, longer gaps lead to blurred representations and diluted depth cues. In contrast, a single-frame baseline maintains coherence with limited drift, explaining its superior and more stable performance.

4.6. Comparative performance analysis

Following the optimization of the auxiliary module's placement and the selection of the most effective temporal structure, the study conducted a detailed comparative analysis to assess the impact of the proposed modifications. The modified versions of VisualDet3D [13] and Monodle [14] were evaluated against their respective baselines, and further comparisons were made with state-of-the-art monocular 3D object detection methods, including CaDDN [35] and Kinematic3D [36]. The evaluation focused on 3D detection performance across different object categories and difficulty levels.

4.6.1. Performance of modified VisualDet3D

Tables 4–6 summarize the performance of the modified VisualDet3D [13] on the KITTI benchmark across car, pedestrian, and cyclist categories. For cars (Table 4), while performance on Easy and Moderate cases remains largely stable, consistent gains are achieved in challenging

scenarios. Notably, AP_{3D} at an IoU threshold of 0.7 improves by 1.94 %, 1.54 %, and 1.05 % for Easy, Moderate, and Hard levels, respectively, with AP_{BEV} exhibiting similar trends, confirming improved spatial localization.

For pedestrians (Table 5), evaluated at an IoU threshold of 0.5, the modified model shows clear improvements in 2D localization and size estimation, particularly in the Easy category (+2.45 % AP_{2D}). Moreover, AP_{BEV} and AP_{3D} consistently improve across all difficulty levels, with gains of approximately 0.5 %–1.07 % and 0.5 %–0.87 %, respectively, indicating enhanced depth perception. Cyclist detection (Table 6) follows a similar pattern, with notable AP_{2D} gains across all difficulty levels and consistent, though smaller, improvements in AP_{BEV} and AP_{3D} . These results demonstrate that temporal feature integration is particularly effective for detecting small and dynamic objects.

Qualitative results in Figs. 9–11 further illustrate these improvements. The modified model successfully detects partially occluded vehicles and cyclists that are missed by the baseline, as well as dynamic objects moving out of the field of view, highlighting the advantage of leveraging temporal context over single-frame inference.

The modified VisualDet3D model was further evaluated on the nuScenes dataset (40,000 annotated samples; 28k/6k/6k for train/validation/test), with results for car and pedestrian categories reported in Tables 7 and 8. For cars, 3D detection accuracy shows no substantial change at a confidence threshold of 0.7, while a clear improvement is observed at a threshold of 0.5, indicating enhanced vehicle localization

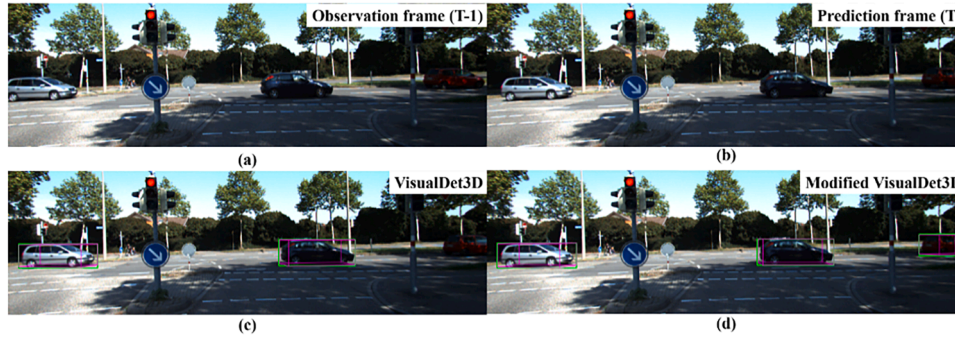


Fig. 11. Visualization result of VisualDet3D and the modified VisualDet3D with the proposed convolutional LSTM Cell auxiliary module.

Table 7

Modified VisualDet3D performance of the car category on the nuScenes test set.

	AP _{2D} (0.7) ↑	AP _{BEV} (0.7) ↑	AP _{3D} (0.7) ↑	AP _{BEV} (0.5) ↑	AP _{3D} (0.5) ↑
VisualDet3D [13]	43.13	0.26	0.05	7.25	4.54
Ours	42.93	0.36	0.07	7.61	5.32
	(−0.2)	(+0.1)	(+0.02)	(+0.36)	(+0.78)

Table 8

Modified VisualDet3D performance of the pedestrian category on the nuScenes test set.

	AP _{2D} (0.7) ↑	AP _{BEV} (0.7) ↑	AP _{3D} (0.7) ↑	AP _{BEV} (0.5) ↑	AP _{3D} (0.5) ↑
VisualDet3D [13]	25.67	0.44	0.16	2.91	2.59
Ours	30.38	0.47	0.37	3.20	2.98
	(+4.71)	(+0.03)	(+0.21)	(+0.29)	(+0.39)

enabled by the proposed auxiliary module. In contrast, pedestrian detection exhibits consistent 3D accuracy gains at both confidence thresholds (0.5 and 0.7), demonstrating improvements in object recognition as well as position and size estimation. Although the proposed module follows an assistive design and cannot fully compensate for severely degraded baseline predictions, the nuScenes results confirm that it consistently improves detection accuracy and localization even when the original VisualDet3D performance is suboptimal.

4.6.2. Performance of modified monodle

In addition to VisualDet3D [13], the proposed auxiliary module was evaluated on Monodle [14], with quantitative results reported in Tables 9–11. For the car category (Table 9), substantial gains are achieved, with AP_{3D} and AP_{BEV} in the Moderate setting improving by 3.94 % and 4.45 %, respectively, indicating more accurate object size and spatial localization.

Pedestrian results (Table 10) reveal a characteristic trade-off. Detection performance on the 2D plane improves for Moderate and Hard cases, while a noticeable decline is observed in the Easy category, accompanied by minor reductions in AP_{BEV} and AP_{3D} across most difficulty levels. This behavior suggests that the temporal module enhances spatiotemporal modeling for challenging pedestrian scenarios involving occlusion or motion but may introduce redundant or noisy features for fully visible, static pedestrians. The emphasis on complex temporal patterns therefore improves robustness in difficult scenes at the expense of slight precision loss in simpler cases.

For cyclists (Table 11), the most pronounced improvements occur in AP_{2D}, with gains of 2.7 %, 0.77 %, and 0.76 % for Easy, Moderate, and Hard levels, respectively. Although small decreases in AP_{BEV} and AP_{3D} are observed, the results indicate more accurate localization of cyclists, particularly in dynamic and cluttered urban environments.

Qualitative comparisons in Figs. 12 and 13 further support these findings. The modified Monodle successfully detects heavily occluded vehicles and tracks objects moving out of the frame, i.e. scenarios where the baseline model fails, demonstrating the effectiveness of temporal

feature integration in enhancing monocular 3D detection under challenging conditions.

4.6.3. Comparison with state-of-the-art methods

Experimental results on VisualDet3D [13] and Monodle [14] confirm that temporal information is particularly beneficial for detecting small and occluded objects. To benchmark the proposed approach, Table 12 compares the modified VisualDet3D and Monodle with MonoCD, CaDDN, Kinematic3D, and other state of the art monocular 3D detectors. While MonoCD [37] achieves near top AP_{3D} performance for cars, particularly in Moderate and Hard settings due to its depth prediction strategy, it does not report results for Pedestrian and Cyclist categories. Similar omissions are observed for other depth assisted methods such as MonoDSSM [38], MonoMAE [39], MonoDFNet [40], and DPL [41], suggesting potential limitations in handling smaller or more complex objects. In contrast, the modified VisualDet3D and Monodle demonstrate broader generalization by achieving competitive or superior performance across all three KITTI object categories, highlighting their practical advantage for real world monocular 3D detection scenarios without relying on additional sensors.

In terms of computational efficiency, inference speed is evaluated on a single NVIDIA RTX A5000 GPU. As shown in Table 12, the proposed temporal auxiliary module introduces only a marginal overhead, reducing inference speed by approximately 1.5 FPS for VisualDet3D and 1.2 FPS for Monodle. Despite this minor cost, the proposed method remains faster than all other compared approaches while maintaining strong 3D detection performance, demonstrating a favorable balance between accuracy and efficiency.

4.6.4. Summary of experimental findings

The experimental results demonstrate that the proposed ConvLSTM-based auxiliary module improves monocular 3D object detection by capturing temporal dependencies when inserted before the prediction head. Temporal feature integration enhances localization accuracy,

Table 9

Modified Monodle performance of the car category on the KITTI test set.

	AP _{2D} (0.7) ↑			AP _{BEV} (0.7) ↑			AP _{3D} (0.7) ↑		
	E	M	H	E	M	H	E	M	H
Monodle [14]	95.54	87.09	78.87	23.74	23.03	21.43	17.26	19.16	16.71
Ours	95.92 (+0.38)	87.37 (+0.28)	79.10 (+0.23)	28.19 (+4.45)	23.49 (+0.46)	21.82 (+0.39)	21.20 (+3.94)	19.77 (+0.61)	16.99 (+0.28)

Table 10

Modified Monodle performance of the pedestrian category on the KITTI test set.

	AP _{2D} (0.5) ↑			AP _{BEV} (0.5) ↑			AP _{3D} (0.5) ↑		
	E	M	H	E	M	H	E	M	H
Monodle [14]	74.38	59.74	51.27	8.94	7.70	6.99	6.90	7.13	5.44
Ours	66.21 (−8.17)	64.13 (+4.39)	56.02 (+4.75)	8.32 (−0.62)	6.52 (−1.18)	6.34 (−0.65)	6.70 (−0.2)	6.03 (−1.1)	5.62 (+0.18)

Table 11

Modified Monodle performance of the cyclist category on the KITTI test set.

	AP _{2D} (0.5) ↑			AP _{BEV} (0.5) ↑			AP _{3D} (0.5) ↑		
	E	M	H	E	M	H	E	M	H
Monodle [14]	67.55	45.55	45.09	8.79	5.48	5.49	7.20	5.40	5.40
Ours	70.25 (+2.7)	46.32 (+0.77)	45.85 (+0.76)	7.96 (−0.83)	5.65 (+0.17)	5.65 (+0.16)	6.51 (−0.69)	5.50 (+0.1)	5.51 (+0.11)

depth estimation, and robustness to occlusion, with particularly clear benefits for small objects such as pedestrians and cyclists. Across both qualitative and quantitative evaluations, the modified VisualDet3D and Monodle consistently outperform their baseline counterparts, achieving notable gains in AP_{3D} and AP_{BEV} across object categories.

The proposed Temporal Deep Feature (TDF) module also exhibits inherent limitations. In simple or static scenes, temporal aggregation can introduce over-smoothing and slightly reduce precision for fully visible objects, while abrupt viewpoint or illumination changes may cause transient instability due to recurrent state reinitialization. In crowded scenes, minor localization drift may occur as a result of feature leakage between adjacent objects. These behaviors are consistent with the modest Easy-category performance decline observed in Table 10 and indicate that temporal cues are most effective in challenging or occluded scenarios, with diminishing returns when temporal information is redundant.

Although the proposed models do not exceed the performance of depth-assisted methods such as MonoCD and CaDDN, they offer a computationally efficient and cost-effective solution for real-world deployment. By leveraging sequential frames through convolutional recurrent modeling, this work advances monocular 3D object detection in scenarios where LiDAR or stereo sensing is unavailable and provides a foundation for future research on self-supervised depth estimation and adaptive temporal modeling.

5. Conclusions

This work presents a generalizable Temporal Deep Feature Learning framework for monocular 3D object detection that leverages convolutional recurrent modeling to exploit temporal dependencies in video sequences. The proposed ConvLSTM-based auxiliary module, MonoTDF, enriches spatial features with temporal context, effectively mitigating depth ambiguity, occlusion, and localization instability inherent to monocular perception. Owing to its model-agnostic design, the module can be seamlessly integrated into existing detectors without architectural modification, as demonstrated on two representative frameworks,

VisualDet3D and Monodle, with consistent performance gains across object categories and metrics on the KITTI and nuScenes datasets.

Experimental results show that temporal feature integration improves both 3D Average Precision and BEV accuracy, particularly for small, distant, or partially occluded objects, while introducing only minimal computational overhead by requiring a single additional previous frame during inference. This efficiency makes the proposed approach suitable for real-time autonomous driving and robotic applications without reliance on depth sensors such as LiDAR or stereo cameras. While this study focuses on KITTI and nuScenes, future work will extend evaluation to larger-scale benchmarks such as Waymo [42] and Argoverse [43] to further assess the generalization capability of the proposed framework.

CRediT authorship contribution statement

Xiu-Zhi Chen: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Yi-Kai Chiu:** Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Chih-Sheng Huang:** Supervision, Project administration, Methodology, Investigation, Funding acquisition. **Yen-Lin Chen:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Yen-Lin Chen reports financial support and article publishing charges were provided by National Science and Technology Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

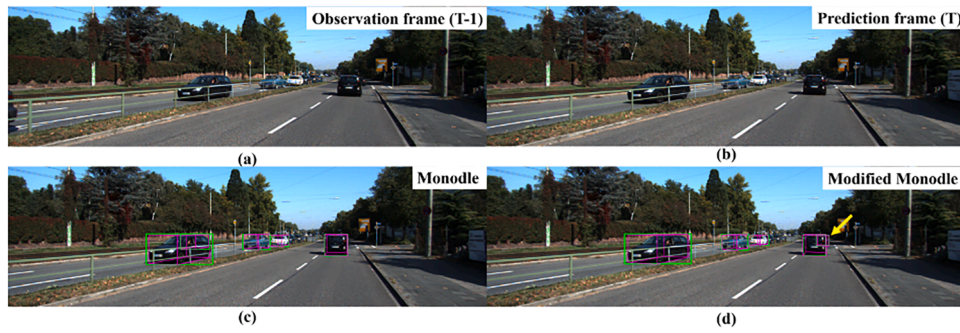


Fig. 12. Visualization result of Monodde and the modified Monodde with the proposed convolutional LSTM Cell auxiliary module.

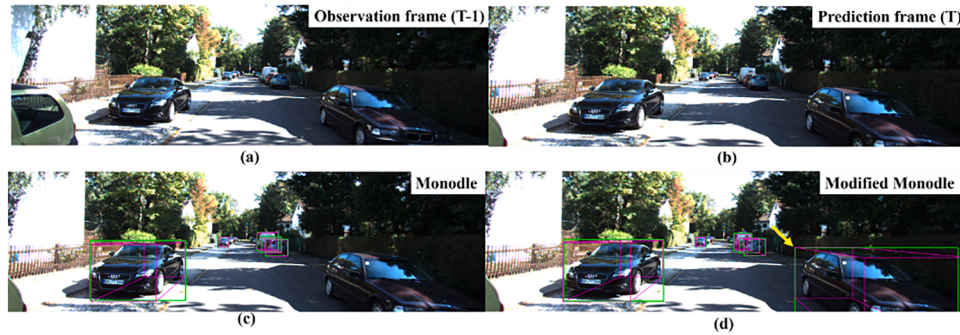


Fig. 13. Visualization result of Monodde and the modified Monodde with the proposed convolutional LSTM Cell auxiliary module.

Table 12

Comparison between modified structure and other 3D object detection methods.

AP3D ↑	Car			Pedestrian			Cyclist			Inference Speed (FPS)
	E	M	H	E	M	H	E	M	H	
MonoCD [37]	25.53	16.59	14.53	-	-	-	-	-	-	19.5
CaDDN [35]	19.17	13.41	11.46	12.87	8.14	6.76	7.00	3.41	3.30	1.6
Kinematic3D [36]	13.01	9.43	7.38	1.19	0.57	0.57	0.00	0.00	0.00	19.7
VisualDet3D [13]	19.43	13.60	10.82	6.94	5.11	4.31	2.44	1.41	1.43	28.7
Monodde [14]	17.26	19.16	16.71	6.90	7.13	5.44	7.20	5.40	5.40	30.1
MonoDSSM [38]	21.47	14.55	11.78	-	-	-	-	-	-	26
MonoMAE [39]	25.60	18.84	16.78	-	-	-	-	-	-	20.5
MonoDFNet [40]	25.71	19.07	15.96	-	-	-	-	-	-	17.6
DPL [41]	24.19	16.67	13.83	-	-	-	-	-	-	20.3
Dp-M3D [44]	23.41	13.65	12.91	8.20	5.75	4.62	5.28	2.96	2.53	22.3
Modified VisualDet3D	21.24	15.78	12.07	7.94	6.08	4.92	4.55	2.15	2.27	27.2
Modified Monodde	21.20	19.77	16.99	6.70	6.03	5.62	6.51	5.50	5.51	28.9

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Data availability

The data that has been used is confidential.

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